

Teacher Answer Key

Part 1 — Sort It Out

- Expressions: $3x + 5$, 12 , $4 + 9$
- Equations: $7 = 7$, $y = 3x$, $2(4x^2 - 9) = 8x^2 - 18$
- Formulas: $A = bh$ (an equation stating a known rule — accept $y = 3x$ with justification)
- Other: $x/2 > 6$ — an inequality (a math sentence whose verb is $>$ instead of $=$)

Big question: the equals sign. Equation = sentence; expression = phrase; “=” is the verb “is.”

Part 2 — Expression Anatomy

- 3 terms: $8x^2$, $7x$, -15
- Coefficients: 8 (of x^2) and 7 (of x)
- Constant: -15 . It is a term: terms are separated by $+$ or $-$, and a term need not contain a variable.
- Middle term of $y^2 - y + 3$ is $-y$, so the coefficient is -1 (the invisible 1).
- 3 terms; the second term has 3 different variables (x , y , z).
- Coefficient of the third term $x^5y^2z^4$ is 1.

Part 3 — Build Your Own (sample answers)

- e.g., $4x^2 + 3x - 7$. Check: 3 terms? coefficient of x^2 is 4? constant -7 ?
- e.g., $2 + 5 = 7$ (true, but states no general rule).
- e.g., $P = 4s$ for the perimeter of a square: P = perimeter, s = side length.

Part 4 — True or False

- TRUE — one expression on each side of the $=$ sign.
- FALSE — a single number is an expression with one term (a constant).
- FALSE — 7 is the coefficient; x is the variable.
- FALSE (subtle!) — the last term is -15 , not 15. The sign travels with the term.
- TRUE — formulas are the equations that state known general rules.

Part 5 — Challenge

- x
- 2 terms: b^2 and $-4ac$
- An equation — it has an $=$ sign joining two expressions (x and the big right-hand side).
- Expected names: variable, coefficients (a , b , c stand in for numbers), terms, expression, equation. Likely “?” items: $\pm$ and $\sqrt{\quad}$ — preview, don't teach.

Part 6 — Question Compass (sample questions about coefficients)

- What: What is a coefficient?
- Example / Non-example: Why isn't the 2 in x^2 a coefficient?

- Reverse: Write an expression where the coefficient of y is -1 .
- Break it: Is there a term with no coefficient? Are you sure? (invisible 1)
- Connect: How do coefficients relate to terms? Can a constant term have one?
- Disguise: How would a test hide a coefficient question inside a harder problem?
- Grade the QUESTIONS, not just answers: variety across directions, precision of wording, fairness of any trap.

Exit Ticket

- Look for: expression = math phrase (no $=$); equation Hi Rob,
- Thank you for taking the time to speak with me this morning. I really enjoyed our conversation, especially understanding the company's pain points and what and what you're looking for in the people you're building the team with.
- Every person I've spoken to sounds really enthusiastic about the job, and that just reinforced my excitement about the opportunity of working in Ascend's technical branch. I'd love to bring my technical and interpersonal skills to the team in helping provide people better healthcare.
- Looking forward to hearing about next steps.
- Best,
- Leo = math sentence saying two things are equal.
- $5y^2 - y + 3 = 0$: equation; left side is an expression with 3 terms ($5y^2$, $-y$, 3); coefficients 5 and -1 ; constant 3; variable y ; right side is the expression 0.